POWER BI RETAIL ANALYTICS IMPLEMENTATION & REPORTING GUIDE

Google Sheets ingestion • Power Query ETL • Semantic modeling • DAX • Report development

Hands-on lab documentation | 4 November 2026

Document purpose

This guide records the implementation evidenced by the supplied Power BI screenshots. It explains the steps, modeling decisions, DAX concepts, visuals, findings, and remaining work. Screenshots are included as evidence and captioned in sequence.

1. Executive summary

This practical exercise builds a retail analytics semantic model and report in Power BI Service (app.powerbi.com), using three related Google Sheets/CSV-style entities: customers, orders, and sales.

The work shown in the screenshots covers connecting to Google Sheets, selecting the three tables, applying Power Query transformations, creating a relational model, adding a calendar table, defining DAX measures, and building initial Sales Overview and Customer Analysis visuals.

The model follows a transaction-oriented structure: customers describe who placed orders, orders describe the order header, and sales contains the individual product/order-line records. This separation is important because one customer can place many orders and one order can contain multiple sales lines.

2. Scope, data entities and grain

customers: one row per customer. customer_id is the key. Attributes include customer segment, city, state, names, and signup_date.

orders: one row per order. order_id is the key; customer_id links each order to its customer. Attributes include order_date, delivery_date, order_status, payment_method and shipping_city.

sales: one row per order-product line. sales_id is the row key; order_id links each line to its order. Attributes include product_id, product_name, product_category, price_band, quantity, unit_price, gross_amount, discount_amount and net_sales_amount.

Grain matters: sales is more detailed than orders. When joining order-level data to line-level sales, order-level amounts or counts can be duplicated. Use DISTINCTCOUNT(orders[order_id]) for order counts, and aggregate line-level amounts from sales.

3. Connecting to the source

The screenshots show the Google Sheets connector in Power Query. A Google Sheet URL and a Google Account credential are selected, followed by the list of available tables: customers, orders and sales.

Select all three entities and inspect the preview before transforming. The initial preview shows generic column labels such as Column1, Column2, etc., while the first row contains the actual field names. This is why promoting headers is an essential early transformation.

Connection note: the screenshots show a Google Sheets connection named e-commerce-gsheet. Credentials and access are tied to the signed-in Google account. For refresh in the Service, ensure the connection remains authorized and that the account has access to the source spreadsheet.

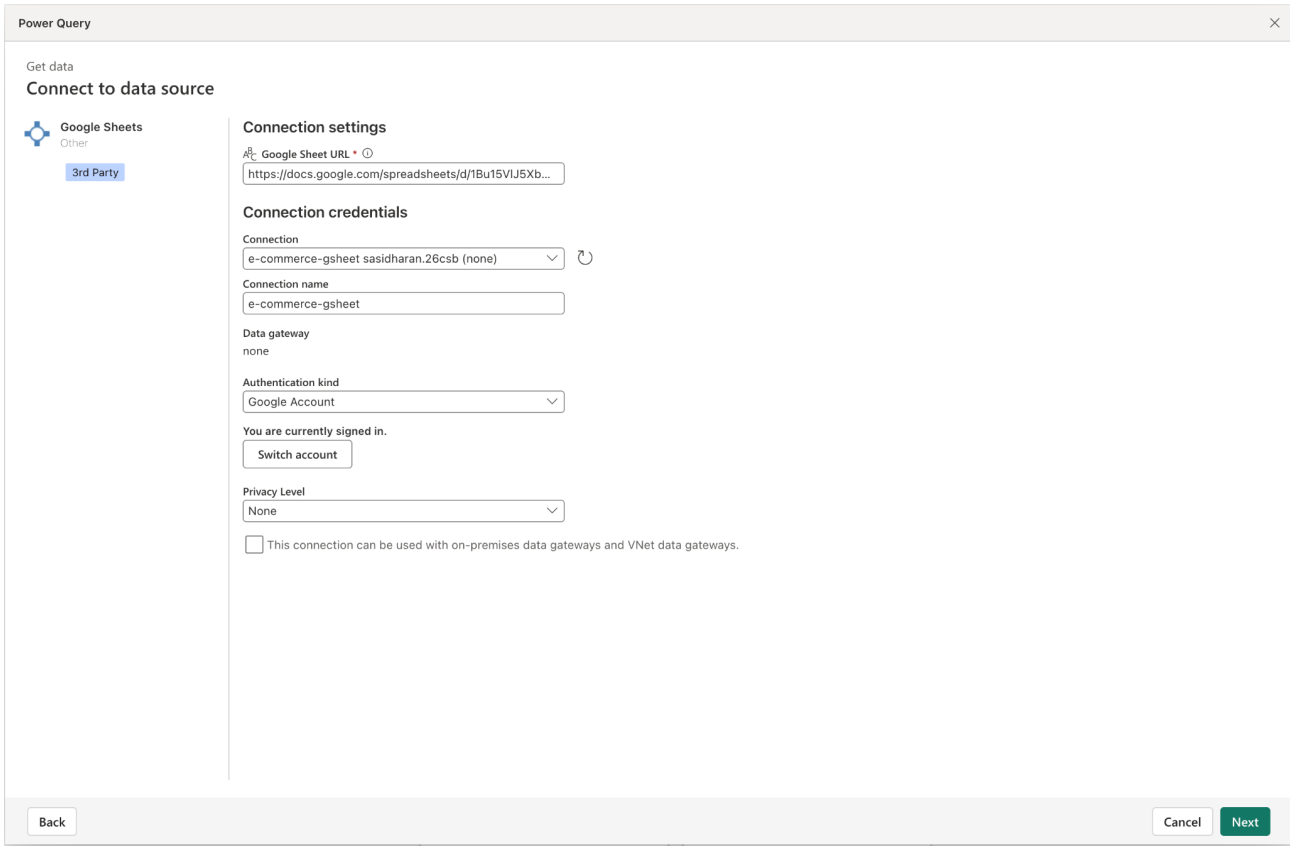
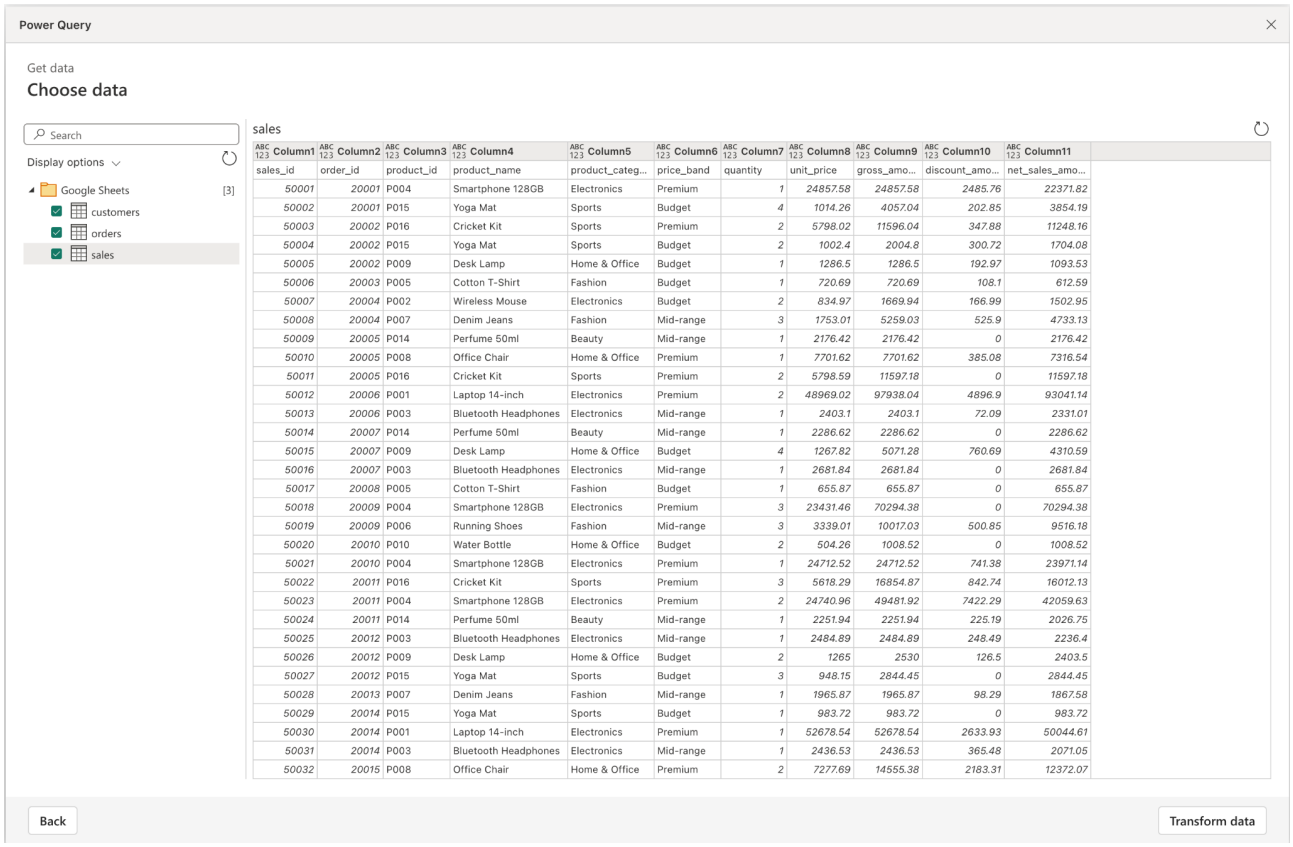
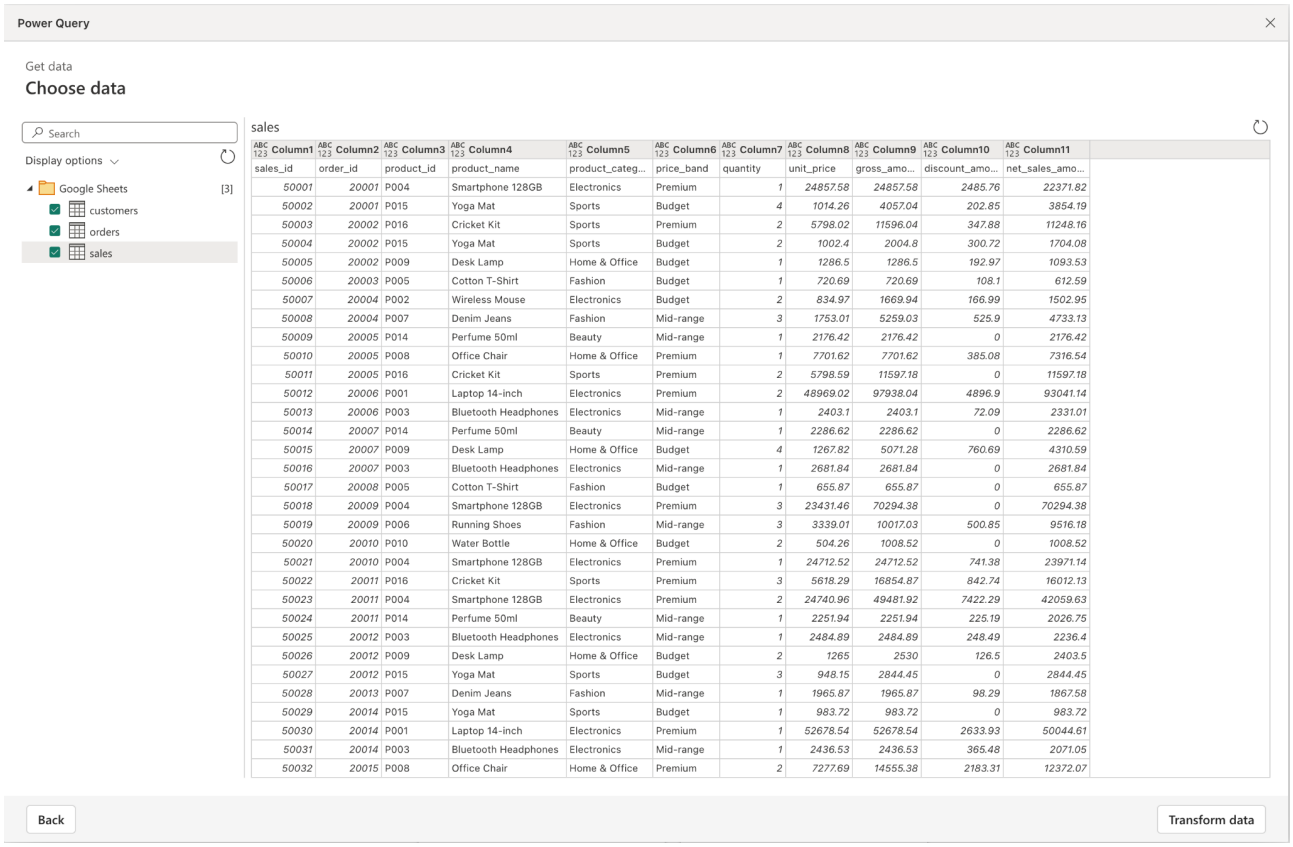


Figure 1. 01 google sheets connection



Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8	Column9	Column10	Column11
sales_id	order_id	product_id	product_name	product_categ...	price_band	quantity	unit_price	gross_amo...	discount_amo...	net_sales_amo...
50001	20001	P004	Smartphone 128GB	Electronics	Premium	1	24857.58	24857.58	2485.76	22371.82
50002	20001	P015	Yoga Mat	Sports	Budget	4	1014.26	4057.04	202.85	3854.19
50003	20002	P016	Cricket Kit	Sports	Premium	2	5798.02	11596.04	347.88	11248.16
50004	20002	P015	Yoga Mat	Sports	Budget	2	1002.4	2004.8	300.72	1704.08
50005	20002	P009	Desk Lamp	Home & Office	Budget	1	1286.5	1286.5	192.97	1093.53
50006	20003	P005	Cotton T-Shirt	Fashion	Budget	1	720.69	720.69	108.1	612.59
50007	20004	P002	Wireless Mouse	Electronics	Budget	2	834.97	1669.94	166.99	1502.95
50008	20004	P007	Denim Jeans	Fashion	Mid-range	3	1753.01	5259.03	525.9	4733.13
50009	20005	P014	Perfume 50ml	Beauty	Mid-range	1	2176.42	2176.42	0	2176.42
50010	20005	P008	Office Chair	Home & Office	Premium	1	7701.62	7701.62	385.08	7316.54
50011	20005	P016	Cricket Kit	Sports	Premium	2	5798.59	11597.18	0	11597.18
50012	20006	P001	Laptop 14-inch	Electronics	Premium	2	48969.02	97938.04	4896.9	93041.14
50013	20006	P003	Bluetooth Headphones	Electronics	Mid-range	1	2403.1	2403.1	72.09	2331.01
50014	20007	P014	Perfume 50ml	Beauty	Mid-range	1	2286.62	2286.62	0	2286.62
50015	20007	P009	Desk Lamp	Home & Office	Budget	4	1267.82	5071.28	760.69	4310.59
50016	20007	P003	Bluetooth Headphones	Electronics	Mid-range	1	2681.84	2681.84	0	2681.84
50017	20008	P005	Cotton T-Shirt	Fashion	Budget	1	655.87	655.87	0	655.87
50018	20009	P004	Smartphone 128GB	Electronics	Premium	3	23431.46	70294.38	0	70294.38
50019	20009	P006	Running Shoes	Fashion	Mid-range	3	3339.01	10017.03	500.85	9516.18
50020	20010	P010	Water Bottle	Home & Office	Budget	2	504.26	1008.52	0	1008.52
50021	20010	P004	Smartphone 128GB	Electronics	Premium	1	24712.52	24712.52	741.38	23971.14
50022	20011	P016	Cricket Kit	Sports	Premium	3	5618.29	16854.87	842.74	16012.13
50023	20011	P004	Smartphone 128GB	Electronics	Premium	2	24740.96	49481.92	7422.29	42059.63
50024	20011	P014	Perfume 50ml	Beauty	Mid-range	1	2251.94	2251.94	225.19	2026.75
50025	20012	P003	Bluetooth Headphones	Electronics	Mid-range	1	2484.89	2484.89	248.49	2236.4
50026	20012	P009	Desk Lamp	Home & Office	Budget	2	1265	2530	126.5	2403.5
50027	20012	P015	Yoga Mat	Sports	Budget	3	948.15	2844.45	0	2844.45
50028	20013	P007	Denim Jeans	Fashion	Mid-range	1	1965.87	1965.87	98.29	1867.58
50029	20014	P015	Yoga Mat	Sports	Budget	1	983.72	983.72	0	983.72
50030	20014	P001	Laptop 14-inch	Electronics	Premium	1	52678.54	52678.54	2633.93	50044.61
50031	20014	P003	Bluetooth Headphones	Electronics	Mid-range	1	2436.53	2436.53	365.48	2071.05
50032	20015	P008	Office Chair	Home & Office	Premium	2	7277.69	14555.38	2183.31	12372.07

Figure 2. 02 choose csv tables



Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8	Column9	Column10	Column11
sales_id	order_id	product_id	product_name	product_categ...	price_band	quantity	unit_price	gross_amo...	discount_amo...	net_sales_amo...
50001	20001	P004	Smartphone 128GB	Electronics	Premium	1	24857.58	24857.58	2485.76	22371.82
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50010	20005	P008	Office Chair	Home & Office	Premium	1	7701.62	7701.62	385.08	7316.54
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50032	20015	P008	Office Chair	Home & Office	Premium	2	7277.69	14555.38	2183.31	12372.07

Figure 3. 03 choose data preview

4. Power Query ETL and data preparation

The Power Query screenshot shows separate queries for customers, orders and sales, with applied steps including Source, Navigation, Promoted headers, Changed column types, text standardization and removal of columns/duplicates where applicable.

Typical ETL actions represented in this pipeline: promote the first row to headers; set IDs and numeric fields to appropriate types; standardize text casing; trim whitespace; clean inconsistent values; remove duplicate records only where the business key should be unique; and retain meaningful nulls such as undelivered delivery dates.

Do not remove blank delivery_date values indiscriminately. A blank may mean the order has not yet been delivered. Treat it as a business state rather than automatically as bad data.

Recommended validation after Close & Apply: customers[customer_id] should be unique; orders[order_id] should be unique; sales[sales_id] should be unique; every orders[customer_id] should match a customer; every sales[order_id] should match an order; quantity should be positive; monetary columns should be numeric; and net_sales_amount should equal gross_amount minus discount_amount, subject to the dataset's rounding rules.

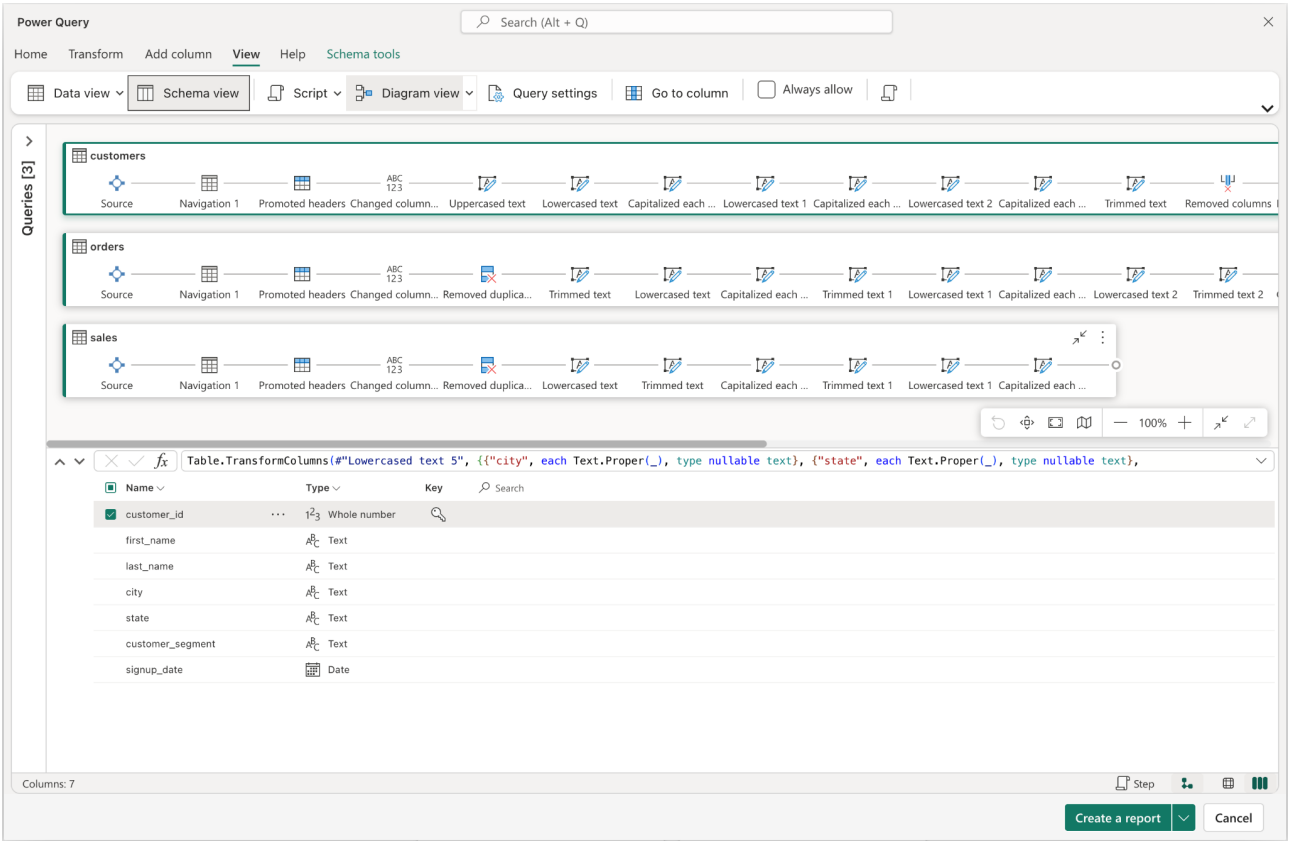


Figure 4. 04 power query transformations

5. Semantic model and relationships

The relationship dialogs show customers[customer_id] to orders[customer_id] configured as One-to-many (1:*), Single cross-filter direction, active. This is correct because each customer can have multiple orders, while each order belongs to one customer.

The second dialog shows orders[order_id] to sales[order_id], also One-to-many (1:*), Single cross-filter direction, active. This is correct because an order can contain multiple product lines.

The resulting model is a simple snowflake-like chain: customers → orders → sales. Filters from customers can flow to orders and then to sales using the single-direction relationships.

Use single-direction filtering by default. Bidirectional filtering can create ambiguous filter paths and unexpected results. Only enable it when a specific modeling requirement is understood and tested.

Important: the sales table is at order-line grain. Total Revenue should sum sales[net_sales_amount]. Total Orders should count distinct order IDs, not rows in sales.

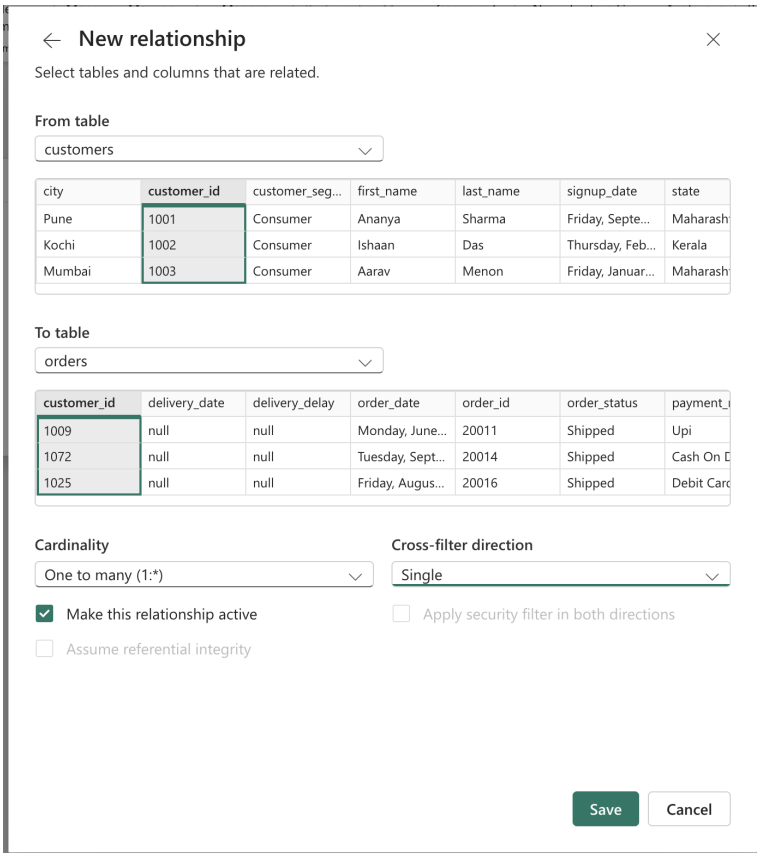


Figure 5. 05 customer orders relationship

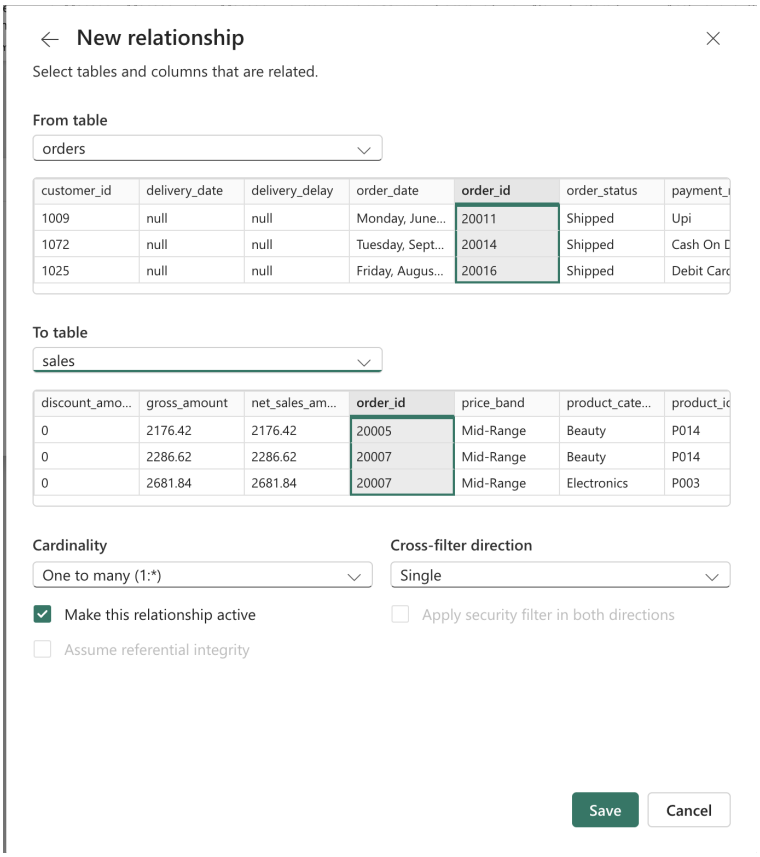


Figure 6. 06 orders sales relationship

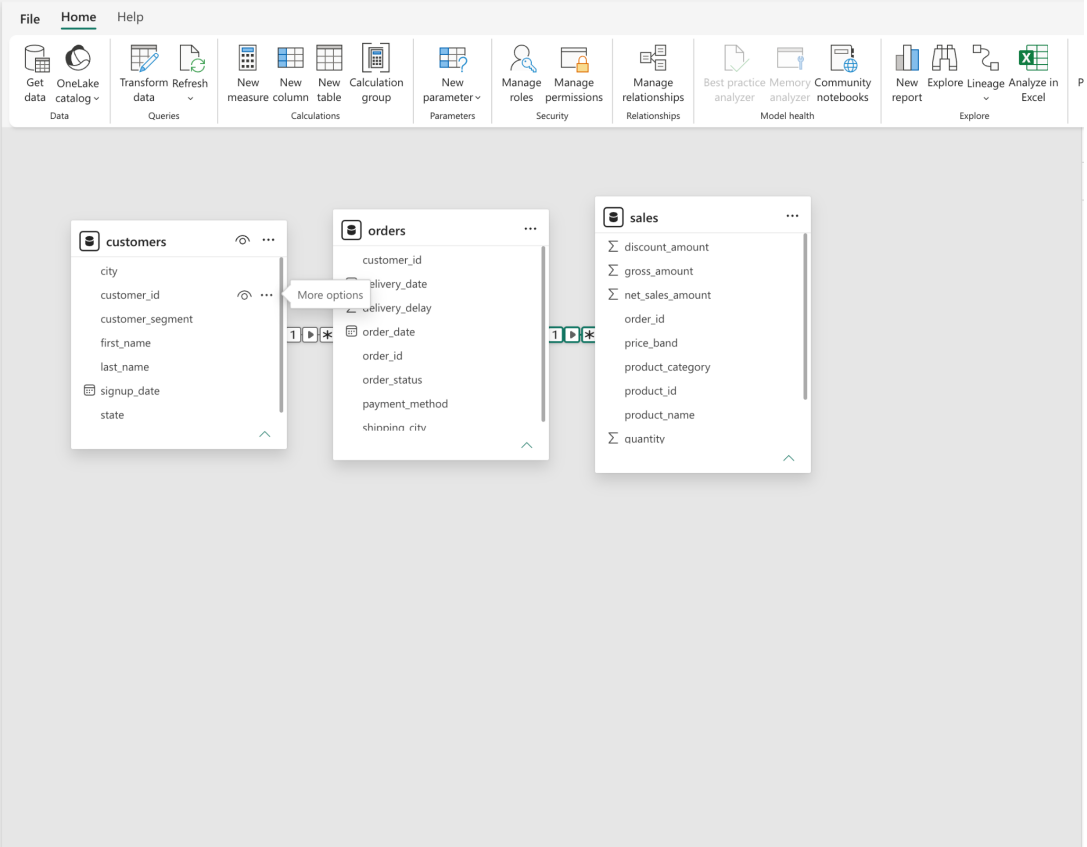


Figure 7. 07 initial model relationships

6. Date table and time intelligence foundation

The screenshots show a calculated DateTable built from the minimum and maximum orders[order_date] values. It adds Year, Month Number, Month, Quarter and Year-Month attributes.

The displayed DAX is:

```
DateTable =
ADDCOLUMNS(
CALENDAR(
MIN(orders[order_date]),
MAX(orders[order_date])
),
"Year", YEAR([Date]),
"Month Number", MONTH([Date]),
"Month", FORMAT([Date], "MMMM"),
"Quarter", "Q" & FORMAT([Date], "Q"),
"Year-Month", FORMAT([Date], "YYYY-MM")
)
```

The relationship dialog shows DateTable[Date] linked to orders[order_date], with one-to-many (1:*), single direction, active. This allows calendar filters to propagate to orders and then sales.

Chronological sorting lesson: Year-Month is a text label, so it may sort alphabetically. The report's monthly chart was corrected to show 2024-01 through 2025-12 in chronological order. A robust model uses a numeric sort key such as YearMonthSort = YEAR(DateTable[Date]) * 100 + MONTH(DateTable[Date]), then selects Year-Month and uses Sort by column → YearMonthSort. Alternatively, use a genuine month-start date column such as DATE(YEAR([Date]), MONTH([Date]), 1) on the axis.

Best practice: mark DateTable as the model's date table using DateTable[Date]. Ensure the calendar has one row per date, no gaps, and covers the full analysis period. For signup-date analysis, use a separate signup-date calendar or an inactive relationship/role-playing date design; the order-date calendar does not automatically filter customers by signup_date.

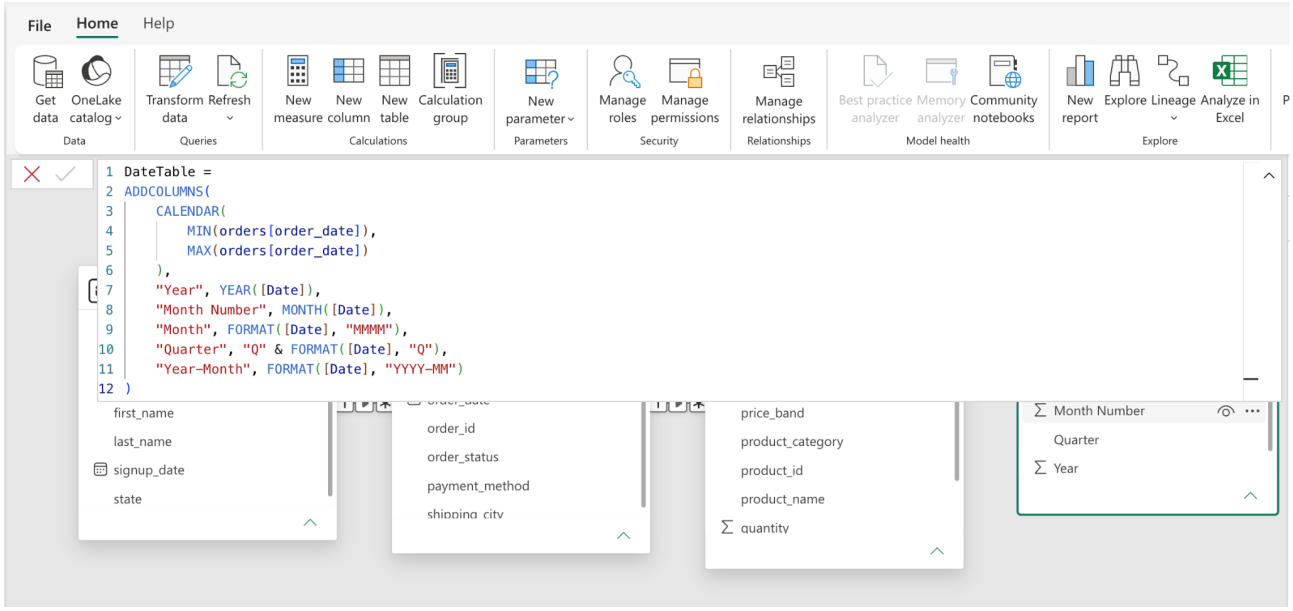


Figure 8. 08 date table dax

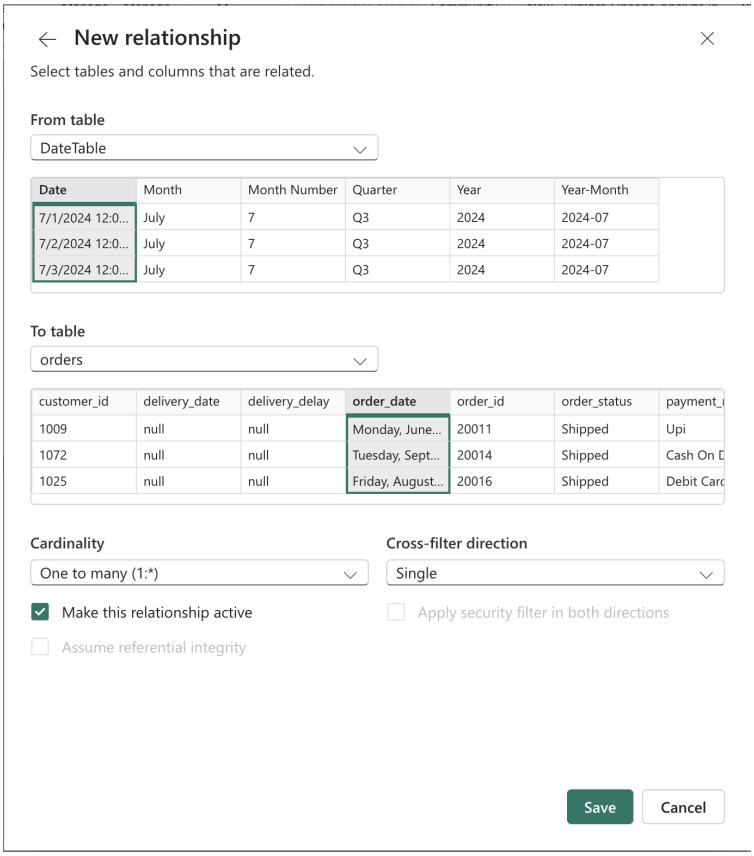


Figure 9. 09 date relationship dialog

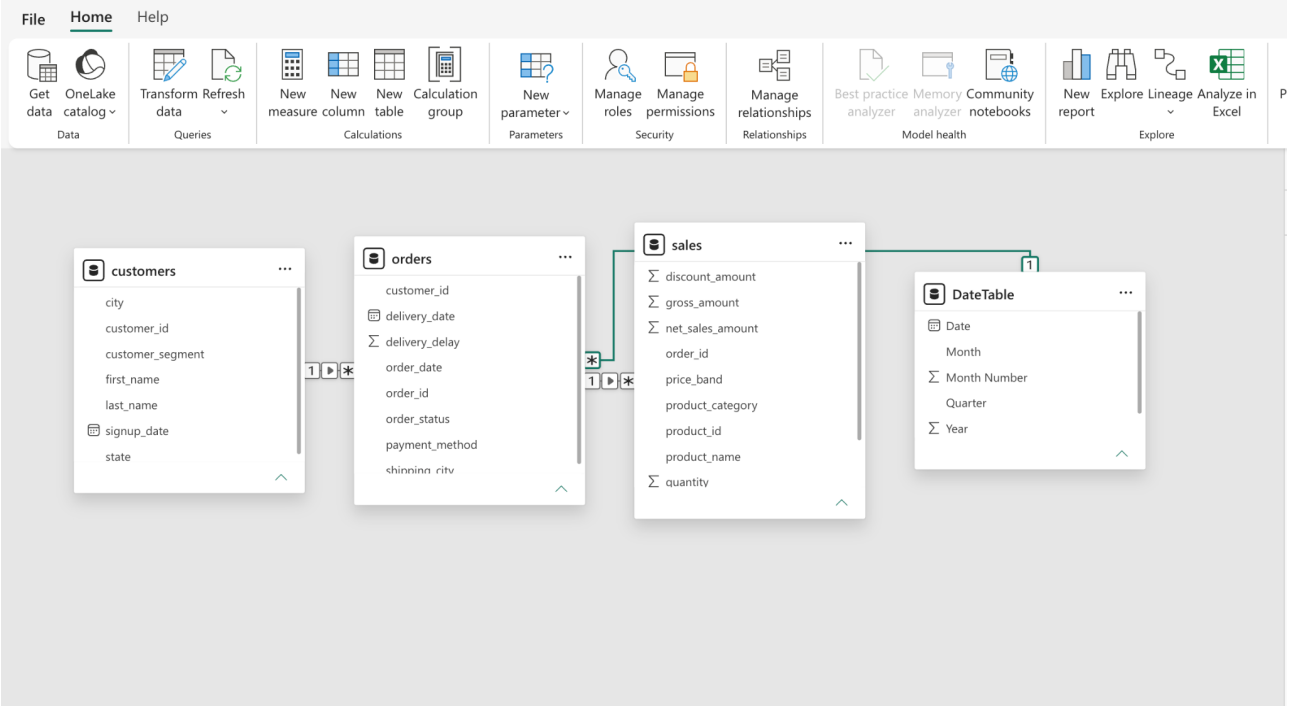


Figure 10. 10 completed model

7. DAX measures created or used

Core measures used in the report:

```
Total Revenue = SUM(sales[net_sales_amount])

Total Orders = DISTINCTCOUNT(orders[order_id])

Total Customers = DISTINCTCOUNT(customers[customer_id])

Units Sold = SUM(sales[quantity])
```

Additional measures recommended for the report:

```
Gross Sales = SUM(sales[gross_amount])

Total Discounts = SUM(sales[discount_amount])

Average Order Value = DIVIDE([Total Revenue], [Total Orders])

Revenue per Customer = DIVIDE([Total Revenue], [Total Customers])

Repeat Customers = COUNTROWS(FILTER(VALUES(customers[customer_id]), CALCULATE(DISTINCTCOUNT(orders[order_id]))
> 1))

Repeat Customer Rate = DIVIDE([Repeat Customers], [Total Customers])
```

Use measures rather than calculated columns for aggregations that should respond dynamically to report filters. Format revenue and monetary measures as currency, counts as whole numbers, and rates as percentages.

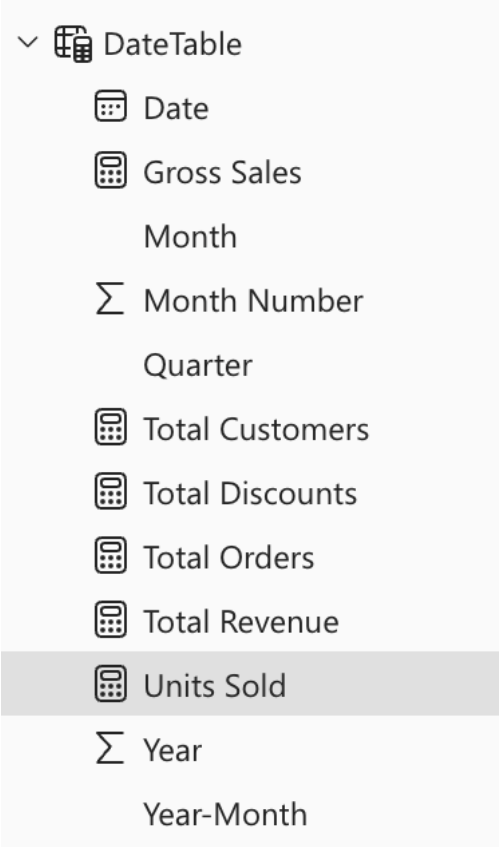


Figure 11. 11 measure list

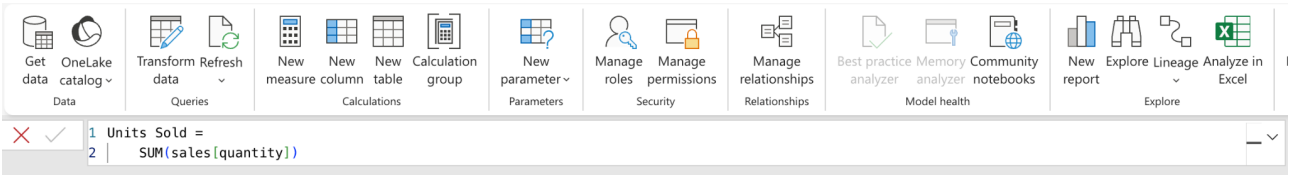


Figure 12. 12 units sold measure

8. Report page 1 — Sales Overview

The Sales Overview screenshot shows the report title RETAIL BUSINESS PERFORMANCE and KPI cards for Total Revenue, Total Orders and Total Customers. The visible values are approximately 5.38M revenue, 260 orders and 80 customers. The fourth KPI, Average Order Value, is a recommended addition but is not visible in the captured page.

The monthly revenue line chart shows a month-by-month series from January 2024 to December 2025. Its chronological axis is an important validation that the Year-Month sorting issue has been resolved.

The Revenue by Product Category bar chart ranks categories. Electronics is substantially larger than the other categories in the screenshot, with Home & Office, Sports, Home & Kitchen, Fashion, Beauty and Books following. The chart uses net revenue from sales lines; it does not measure profit because product costs are not present.

The Monthly Order Volume column chart shows distinct order counts by month across the same 24-month period. It allows comparison of order frequency with revenue: a month with high revenue need not have the highest order count if average order value differs.

Recommended remaining Sales Overview visual: Order Status Distribution using a donut chart with Legend = orders[order_status] and Values = [Total Orders]. Ensure statuses are standardized and decide how cancelled/returned orders should be interpreted in business KPIs.

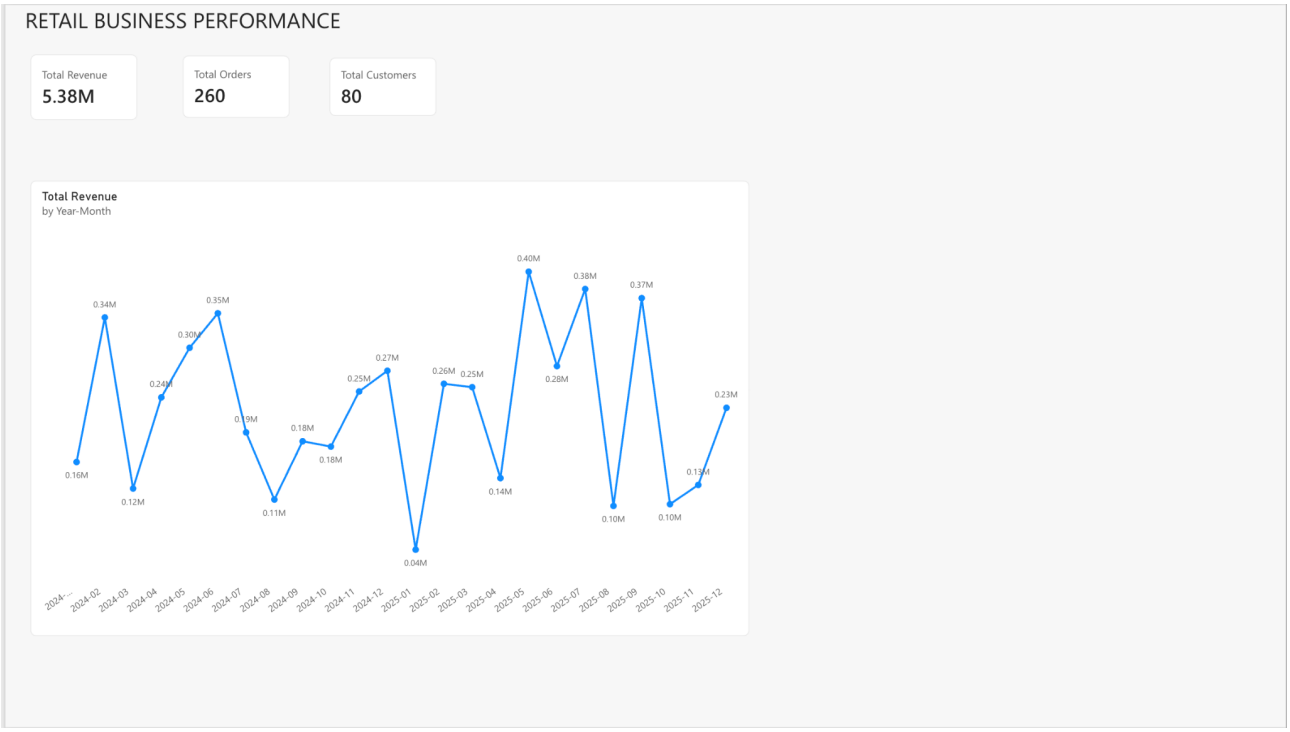


Figure 13. 13 sales overview report

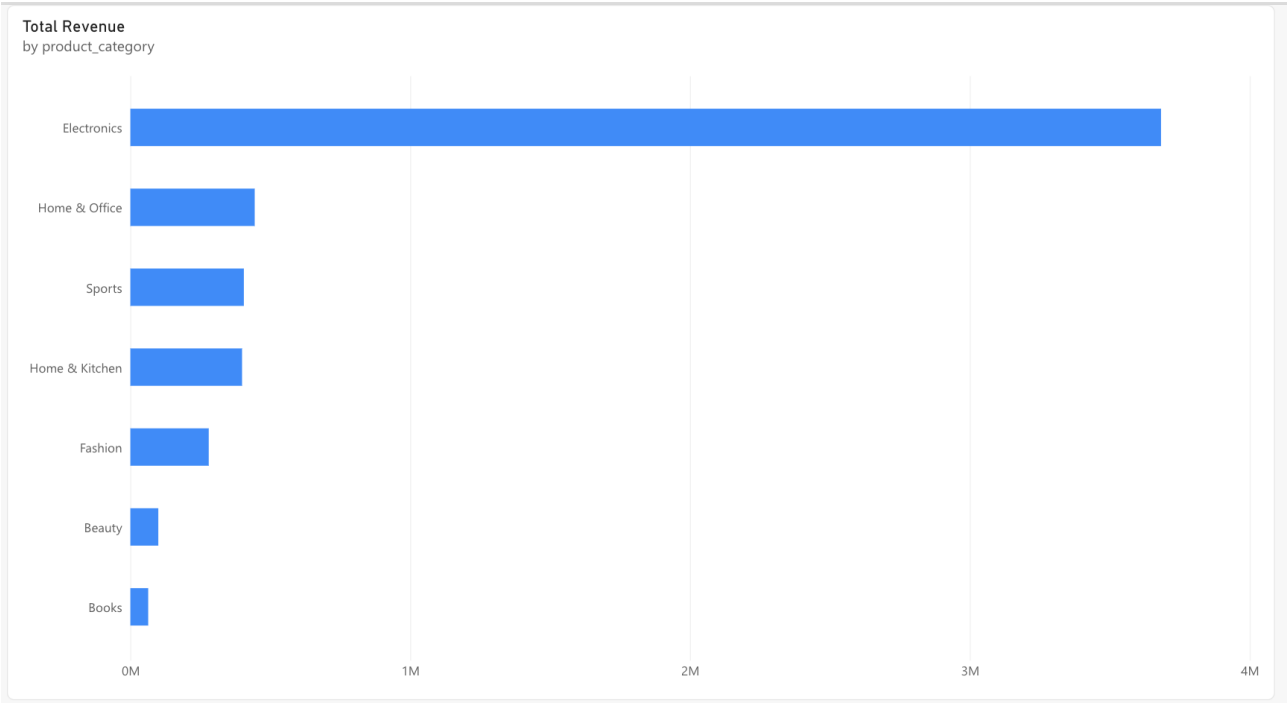


Figure 15. 15 revenue by category

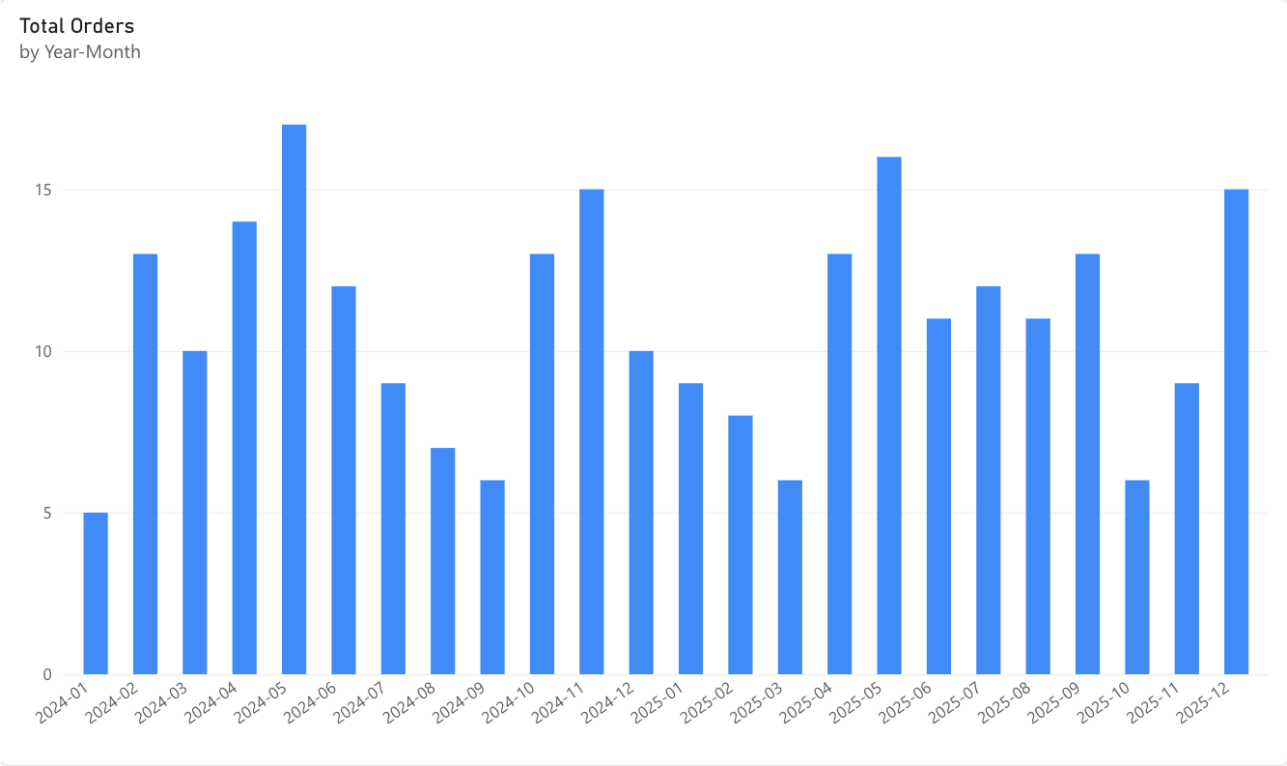


Figure 16. 16 monthly order volume

9. Report page 2 — Customer Analysis

The Customer Analysis screenshot contains four visuals: customer count by customer_segment (donut), new customers by year (bar chart), revenue by customer_segment (bar chart), and total customers by city (bar chart).

The donut shows 80 customers in total, split into Consumer 62 (77.5%), Small Business 12 (15%), and Corporate 6 (7.5%). Consumer is therefore the dominant segment by customer count in this dataset.

The revenue-by-segment chart shows Consumer as the largest revenue contributor, followed by Small Business and Corporate. The difference in revenue is large, so consider whether the segment mix, purchasing frequency or order value explains the result. The current chart alone cannot establish causation.

The city chart shows customer counts by city, with Mumbai highest in the screenshot (12), followed by Bengaluru (10), then Delhi, Kochi and Pune (9 each), Coimbatore (8), Jaipur (7), Chennai and Hyderabad (6 each), and Kolkata (4). This is a geographic distribution of customers, not revenue by city.

The new-customer chart is grouped by signup year and displays 2023, 2024 and 2025. The visual is a year-level view; for a useful acquisition trend, change the axis to month/year-month using a signup-date calendar or an appropriately configured signup_date relationship.

Recommended additions: KPI cards for Total Customers, Repeat Customers, Repeat Customer Rate and Revenue per Customer. Consider a top-N city view if the report becomes crowded.

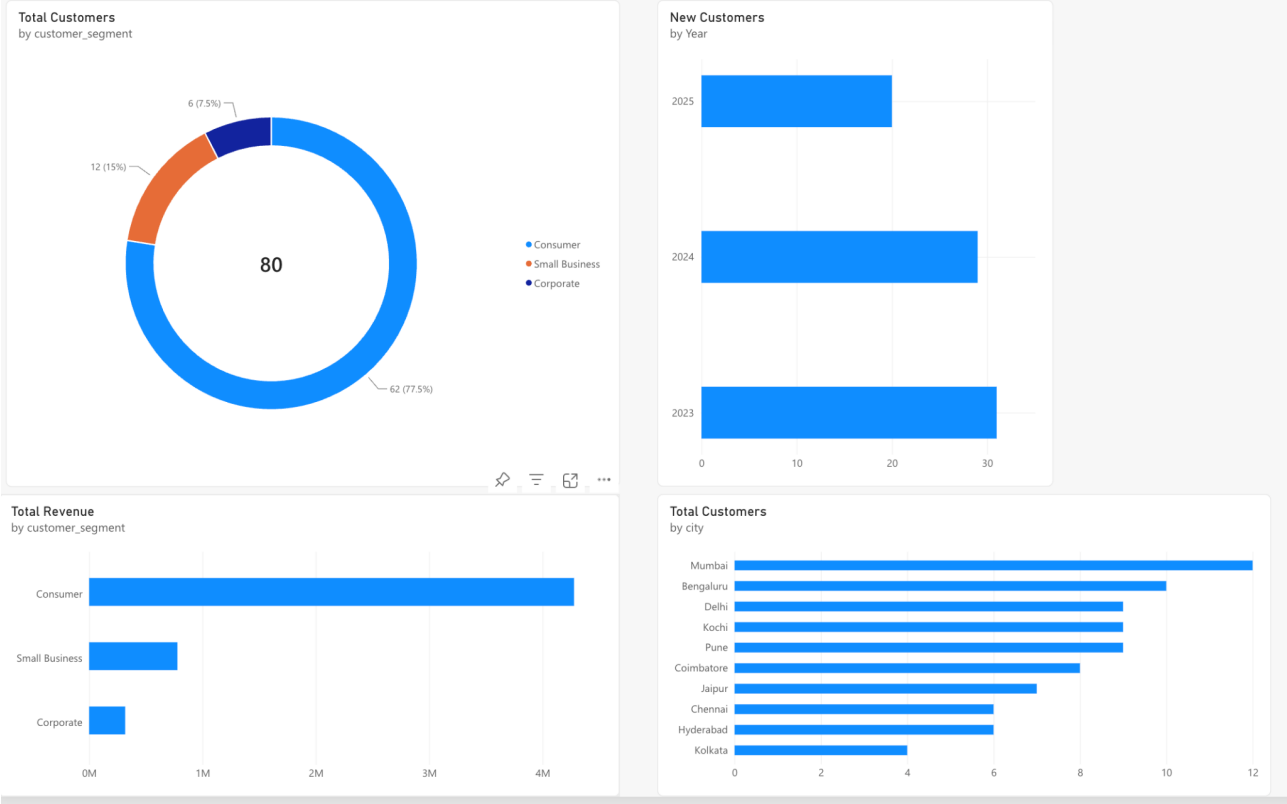


Figure 14. 14 customer analysis report

10. Findings and business reporting

Finding 1 — Electronics dominates category revenue. Evidence: the category bar chart has a much longer Electronics bar than all other categories. Interpretation: the business is heavily reliant on this category for revenue. Follow-up: investigate product-level concentration, stock availability, discounting and supplier dependency. Limitation: revenue is not profit.

Finding 2 — Consumer customers dominate the customer base. Evidence: 62 of 80 customers (77.5%) are Consumer. Interpretation: aggregate results are likely strongly influenced by Consumer behaviour. Follow-up: compare average order value, repeat rate and order frequency by segment.

Finding 3 — Consumer segment also contributes the most revenue. Evidence: Consumer has the longest revenue bar in the segment chart. Interpretation: its revenue leadership may reflect its larger customer count, higher spend per customer, or both. Follow-up: compare revenue per customer and orders per customer by segment.

Finding 4 — Customer distribution varies by city. Evidence: Mumbai has the largest customer count in the displayed city chart. Interpretation: Mumbai may be a useful market for targeted campaigns, but customer count alone does not show market potential or profitability. Follow-up: analyze revenue and repeat purchasing by city.

Finding 5 — Monthly revenue and order volume fluctuate. Evidence: both time-series visuals show substantial month-to-month variation. Interpretation: demand may be uneven; investigate seasonality, campaigns, product mix, and average order value. Limitation: two years of synthetic practice data is not enough to establish a reliable long-term seasonal pattern.

11. Validation, caveats and improvement backlog

Validate the model: confirm unique primary keys; check unmatched foreign keys; verify relationship direction and active status; confirm date types; verify the DateTable covers all order dates; check the Year-Month axis order; and reconcile total revenue to a source-level calculation.

Validate measures: Total Orders should be 260 and Total Customers should be 80 in the unfiltered report as shown in the screenshot. Total Revenue appears as 5.38M in the overview card. Reconcile the exact unrounded value with SUM(sales[net_sales_amount]).

Business definition caveats: decide whether cancelled and returned orders count in Total Orders, Revenue and customer repeat metrics. If returned orders remain included in net sales, the revenue measure may overstate realized sales. Document the chosen policy.

Data caveats: the dataset is synthetic practice data. It should not be treated as evidence of actual market performance. Revenue has no cost-of-goods-sold or operating-cost context, so gross margin/profit cannot be calculated.

Visual improvements: add data labels where they improve readability; use consistent currency formatting; give charts descriptive titles; avoid overloading the canvas; use slicers for year, customer segment and product category; test slicer interactions; and provide tooltips with exact values.

Model improvements: hide technical sort columns from report view; format measures; organize measures in a dedicated measure table; mark DateTable as a date table; add a signup-date calendar for acquisition analysis; and consider a proper product dimension if product attributes expand.

12. Suggested next steps

Complete the Sales Overview page by adding the order-status donut and the Average Order Value KPI. Confirm the monthly order chart is sorted chronologically.

Finish Customer Analysis with the four KPI cards and change new-customer acquisition from annual to monthly granularity if the data supports it.

Build Page 3 — Product Performance: revenue by product, units sold by category, discount rate by category, and price-band performance.

Build Page 4 — Order Analysis: order status, payment method, delivery delay, and order volume trends.

Use DAX Query View to validate grouped totals and Performance Analyzer to identify expensive report visuals. Document before/after results if optimization is attempted.